

Review

Maize and soybean yield prediction using machine learning methods: a systematic literature review

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Abstract

Today's agronomy is data-rich, and machine learning (ML) provides the ability to efficiently predict crop yields, utilizing high-volume data to optimize agricultural decision-making. Numerous ML models are used, yet systemized framework guiding the crop-targeted selection of models, features, accuracy measures, and addressing associated challenges is lacking, specifically for soybean and maize, world's vital crops. Henceforth, this is the first crop-targeted systematic literature review (SLR) performed to retrieve/consolidate the ML techniques and key features utilized in maize and soybean yield prediction research. The study searched four databases (ProQuest, Wiley, Science Direct, and EBSCOhost), producing 1859 articles, which finally reduced to 82 articles following SLR's inclusion/exclusion criteria. These papers were thoroughly analysed for generating common consensus and future research recommendations. Results revealed the temperature, precipitation, historical crop yield, normalized difference vegetation index (NDVI), and soil pH to be the most utilized ML features for yield prediction research. The Random Forest (RF), Artificial Neural Networks (ANNs), Support Vector Machines (SVMs), and Extreme Gradient Boosting (XG-Boost) were identified as the mostly used ML algorithms. Most often applied deep learning techniques include long short-term memory (LSTM) and convolutional neural networks (CNNs). In the utilized models, the most used performance assessment measures were noted as the coefficient of determination (R^2), root absolute error (RAE), root mean square error (RMSE), and mean absolute error (MAE). This SLR also reported major challenges with obtaining high quantity/quality data, model complexity tackling, and incorporating the inclusion of farm management factors in yield prediction process. Altogether, this SLR offers a valuable framework for model selection, feature unification, accuracy measures comparison, model performance assessment, and addressing challenges in ML-based yield prediction research, with an emphasis on maize and soybean.

Keywords Machine learning in agronomy · Deep learning · Crop management · Precision agriculture · Artificial intelligence

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Abbreviations

ML	Machine learning
DL	Deep learning
MLR	Multiple linear regression
ANNs	Artificial neural networks
NNs	Neural networks
SVMs	Support vector machines
SVR	Support vector regression
RF	Random forest
kNN	K nearest neighbours
DNN	Deep neural networks
CNNs	Convolutional neural networks
XGBoost	Extreme gradient boost
GPR	Gaussian process regression
LSTM	Long short-term neural networks
RNN	Recurrent neural networks
EVI	Enhanced vegetation index
NDVI	Normalized difference vegetation index

1 Introduction

Agriculture impacts global food security and economic stability [1]. Maize and soybean are world's two primary crops, with maize being the most consumed cereal (208.23 million hectare) and soybean being the most cultivated legume (136.90 million hectare) [2, 3]. Timely and accurate crop yield predictions are important in improving on-ground decision making in crop production systems for sustainable food production [4]. ML methods have proven to be an effective tool in predicting crop yield especially when remotely sensed datasets are coupled with climate and soil attributes [5]. Unlike traditional statistical modelling-based yield estimation, ML based crop yield prediction is time-and-cost effective and can be performed in-season [6]. The ML applications are successful due to their ability to learn intricate and non-linear relations amongst variables from a high-volume database [5]. Numerous ML algorithms, ranging from the simplest linear regression to ensemble models and deep neural networks, are available for crop yield prediction [7, 8]. The accuracy of ML techniques relies on input variables, number of years of data, training datasets, implementation strategies, and intricacy of research problems [9]. Performance rating of multiple ML model comparisons is a yardstick to find the appropriate ML model for precisely predicting the target variable under question [10]. Formulating a high accuracy prediction model mandates meticulous selection of correct algorithms that can tackle data type/volume relevant to the research problem [7].

Crop yield prediction is an arduous task due to immensely intricate nature of agriculture practices with high degrees of variabilities caused by environment, management, and genetical factors [11]. Moreover, these variabilities alter by region, crop type, and crop development stage necessitating the need of devising tailored approaches in selecting the suitable models customized to specific crops [4]. The utilization of ML in agriculture research started in 1980's [12]. Undoubtedly, ML is increasingly becoming popular in agriculture [13]. However, majority of the ML-agriculture research has, until now, been restricted to applying a specific ML model on a particular crop, typically for simple computation of the impact of inputs on yield [14]. Beyond this, the research on comparing the accuracies of different ML models for specific crops is a work-in-progress. Taking all such studies together for generating a common consensus on crop-targeted ML usage is lacking [7]. This has taken away the research abilities to generalize the applicability of different ML models and assess the transferability of their applications in diverse cropping systems in agriculture [8]. Considering ML's increasing role in agricultural research over the last four decades, literature should have a consensus on ML-yield prediction research via meta-analysis, narrative, or SLR. The required review needs to focus on key aspects such as identifying the suitable ML models and outlining the ideal features critical to improved accuracy in context of yield prediction [7]. Notably, current literature is lacking such review studies on crop yield prediction, with even fewer studies specifically focused on individual crops including maize and soybean [8]. Henceforth, to address this research gap, we carried out a SLR which is the first attempt in reviewing the ML and DL methods and assessing their accuracy of yield prediction for maize and soybean crops. This can aid future researchers in decision making for model selection. A successful SLR must be comprehensive,

systematic, and easily replicable [15]. Therefore, this paper covered entire literature that has published to date on ML applications in maize and soybean yield prediction. Furthermore, a clear explanation on the step-by-step workflow was provided for facilitating future researchers in replicating similar studies. This SLR starts by explaining the related work or background of different ML and DL techniques in crop yield prediction. The rest of the paper is categorized in various sections: Sect. 2 explains the methodology; Sect. 3 discusses the results that (i) identified the most used ML and DL techniques, (ii) highlighted the most usable input parameters for the research domain under question, (iii) ascertained the most used type of accuracy measures, (iv) compared the accuracy measures used by different ML/DL/ensemble models to quantify the model performances, and (v) discussed the challenges in research mentioned in the reviewed literature; Sect. 4 presents the discussion on the results; and lastly, Sect. 5 covers the conclusions and future suggestions.

2 Research methodology

The present SLR is conducted in accordance with the basic guidelines suggested by Tranfield et al. [16]. A five-step method displayed in Fig. 1 was followed to meet the needs of our research, which includes pilot search during the first phase to analyze the related literature, followed by deriving the research question and subsequent steps.

2.1 Identification of research questions

Forming a research topic, which leads to the selection of research techniques and procedures, is the most significant aspect of the SLR study design. To analyse the selected literature from several dimensions, following research questions (R-Q)'s has been defined in the first step of systematic investigation:

- R-Q1: Which **ML and DL techniques** have been most widely used to predict the maize and soybean yield?
- R-Q2: What **type of input parameters** have been most extensively used for predicting maize and soybean yield using ML?
- R-Q3: What **type of accuracy measures and software** have been most used for ML and DL techniques for maize and soybean yield prediction?
- R-Q4: What is the **comparative performance and criteria of ML and DL techniques** in the field of crop yield prediction?
- R-Q5: What are the **challenges** in maize and soybean yield prediction research utilizing ML and DL techniques?

2.2 Pilot search, identification of keywords and search strategy

The basic search was conducted using a pilot search method based on a starting input ("machine learning" AND "yield prediction" AND "corn" OR "soybean" OR "maize") in various databases, including ProQuest, Wiley, Science direct, and EBSCOhost. The databases were further scrutinized following the aforesaid string, the papers focused on full text, and peer reviewed scholarly journals, with the customized search for the time span of the research articles published since 2013. The decision of selecting a study period starting from 2013 was informed by the consensus derived from the results of search strings (Table 1) showing that the ML exploration in crop yield prediction initiated in 2013 onwards [17]. The initial pilot search enabled us to finalize the database of the papers based on relevancy, recency, and direct applicability to the research question. Relevancy was assessed by ensuring the alignment of included papers with SLR objectives, through title, abstract, and keywords skimming. For example, some papers were ML, DL, or yield prediction based but

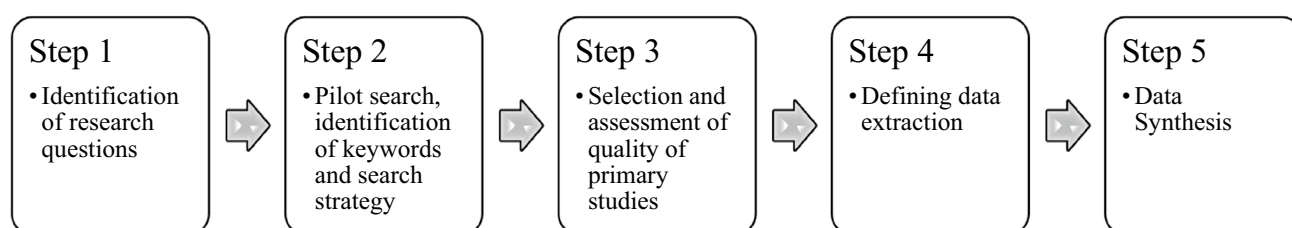


Fig. 1 Framework followed for systematic literature research

Table 1 Final strings for both the databases

Database	Keywords and string
ProQuest	("Machine learning" OR "data mining" OR "deep learning") AND ("yield prediction" OR "yield forecasting" OR "yield estimation") AND ("corn" OR "maize" OR "soybean" OR "soyabean")
Science direct	("Machine learning" OR "data mining" OR "deep learning") AND ("yield prediction" OR "yield forecasting" OR "yield estimation") AND ("corn" OR "maize" OR "soybean" OR "soyabean") (Title, abstract, keywords)
Wiley	[("Machine learning" OR "data mining" OR "deep learning") AND ("yield prediction" OR "yield forecasting" OR "yield estimation") AND ("corn" OR "maize" OR "soybean" OR "soyabean" OR "soyabeen")] (anywhere)
EBSCOhost	("Machine learning" OR "data mining" OR "deep learning") AND ("yield prediction" OR "yield forecasting" OR "yield estimation") AND ("corn" OR "maize" OR "soybean" OR "soyabean" OR "soyabeen") (anywhere)

were related to crops other than maize and soybean. Similarly, some papers were ML, DL, or maize/soybean based, and they explored areas such as crop physiology, abiotic/biotic stresses, or agronomy but were unrelated to yield prediction. All such papers were excluded owing to their non-relevancy to the scope of this SLR. To ensure recency of this SLR in presenting the latest developments in research, all the studies published until 2022 (as the literature search was conducted in 2022) were included. Applicability was the other aspect that was considered, for ensuring that the findings of the included studies should be generalized or matched to the scope of the questions to be answered in this SLR. Thereafter, the initial pilot search with the primary string enabled us to finalize the final string for identifying the relevant and most appropriate research papers from each database. Thus, extracted details of the research papers considering the journal names, author names, publication year, study titles, and abstract skimming was cross examined, and the duplication of the similar studies were removed. This exercise enabled us to identify and reach the final strings containing the most relevant papers. The list of maximum relevant papers was retrieved by the knowledge base available in the four databases used. During the process of string search, we identified maximum keywords satisfying the present research question and developed strings for all databases. Table 1 describes the keywords, and the search strings used thereafter.

2.3 Selection and quality assessment of primary studies

After finalizing the search string, research papers were extracted from the databases using the string. Total 1859 papers were retrieved initially (Table 3). Each research paper was further studied and examined based on the parameters decided for the study, *i.e.*, ML techniques/DL techniques, maize yield prediction, and soybean yield prediction. The database was further trimmed to 908 papers after contemplating only full text and peer reviewed journals that were available in publicly available content database, for the time span *i.e.*, 2013–2022. In the next step, the process of inclusion and exclusion of several parameters was performed where in the journals not dealing with the subject directly or indirectly and in language other than English were excluded and the document type was fixed as published research articles only. A total of 489 studies were finalized after considering the inclusion and exclusion criteria (Table 2). All these studies were further gauged with respect to their title, abstracts, findings, article type, and future research scope, which enabled us to explore and identify the further exclusion of studies. At this stage, only 82 relevant studies were restored from the sample repository for the final analysis.

Table 2 Inclusion and exclusion criteria

Inclusion	Duration	10 years (2013 onwards)
	Language	English
	Publication title	Publicly available content database
Exclusion	Criteria 1	Papers not directly related to maize and soybean yield prediction with machine or deep learning techniques
	Criteria 2	Duplicate retrieved studies from other databases
	Criteria 3	Full text not available and not peer-reviewed
	Criteria 4	Review papers, survey papers, and case studies
	Criteria 5	Subjects Exclusion: Papers related to the subjects such as "unmanned aerial vehicles" OR "sensors" OR "wheat" OR "satellites" OR "mapping" OR "image processing" OR "precision farming" OR "satellite imagery" OR "landsat satellites" OR "phenology" OR "physiology" OR "spatial resolution", but beyond the relevancy to the scope of the objectives of the questions to be answered in the SLR

Table 3 Allocation of extracted papers from the selected databases

Database	Extracted papers	Papers after exclusion criteria	Percentage of papers
ProQuest	1206	61	74.39
Science direct	563	17	20.73
Wiley	85	3	3.66
EBSCOhost	5	1	1.22
Total	1859	82	100

Table 3 is showing the number of papers initially extracted and the number of papers considered after applying the selection criteria.

2.4 Data extraction

The entire data was transformed into a Microsoft excel sheet to avoid potential biases (if any) in the process of SLR. The information was systematically extracted and sorted in categories such as publication year, authors/co-authors, study title, country of research, input parameters used, ML or DL techniques employed, and the evaluation parameters and results.

2.5 Data synthesis

To answer the research questions, the extracted data from the selected studies have been synthesized. Figure 2 represents the stages of PRISMA for the selection of the studies for literature review [18].

The PRISMA stands for Preferred Reporting Items for Systematic Reviews and Meta-analysis. This methodology primarily focuses on reporting of systematic reviews. Although it is not a quality assessment methodology, but useful for critical appraisal of published systematic reviews [18]. Also, it is important to mention here that every research or methodology has its unique set of limitations. Even though this SLR has followed the rigorous methodology, as suggested by previously published SLRs, to search/analyse the literature/findings, there can be the scope of different type of biases with this study. Firstly, excluding the non-English studies may cause geographically biased presentation of research neglecting the key inferences from non-English regions. Secondly, overinclusion of studies from selected regions, context, or population can negatively impact global generalizability of the SLR. Thirdly, reviewer's personal preferences, presumptions, or subjectivity in evaluating the studies can hamper the consistency and objectivity of the review procedures. The results of the publication's analysis are presented in Sect. 3.

3 Results

The year-wise distribution of finally considered papers is displayed in Fig. 3 with the number of published articles peaked in the year 2021. The paper's percentage (2, 4, 2, 1, 2, and 4%) from years 2013 to 2018 signifies that the adoption of ML techniques in maize and soybean yield prediction research has been consistently growing but at moderate rate (Fig. 3). An 84% of the total papers in the research domain published only after 2019. This implies that ML is gaining popularity in the studied domain, with a significant increase in adoption after 2019, indicating a substantial shift from its early phases.

Table 4 presents the listing of articles utilized ML and DL that were finally selected following the inclusion and exclusion criteria for SLR process, and Table 5 shows the listing of selected articles utilized ensemble/hybrid techniques. These tables also show the region and author/year details of selected studies, as well as the input parameters considered for model building.

R-Q1: Which ML and DL techniques have been most widely used to predict the maize and soybean yield?

Current study found that RF was the most popular ML technique in the domain of maize and soybean yield prediction followed by ANNs, SVMs, and XGBoost techniques (Fig. 4). Altogether, these four ML techniques used 59% of the times the total usage frequency amongst all 87 models reported in the reviewed papers. In specific, RF, ANNs, SVMs, and XGBoost were found used 23, 14, 14, and 8%, respectively of the total usage frequency of all 87 models (Fig. 4). The reasoning behind the most broader usability of RF can be attributed to (i) its simplicity in implementation and interpretability, without needing immense hyperparameter tuning, (ii) its ability of effectively handle both quantitative and

Fig. 2 PRISMA flow diagram-different stages for identification, screening, and inclusion of final set of papers considered for the literature review analysis

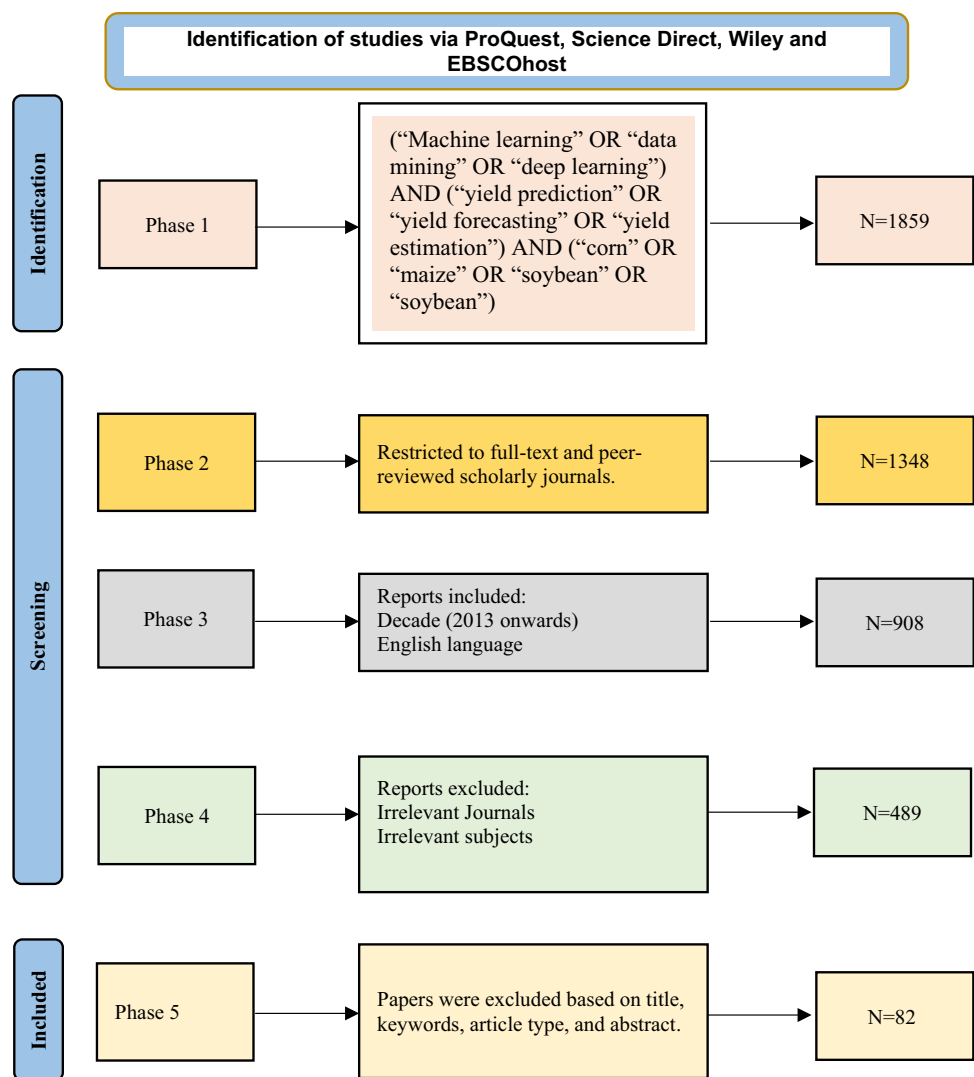


Fig. 3 The annual count and corresponding percentage of scientific articles published in the domain of machine learning applications for predicting maize and soybean yields from 2013 to 2022, derived from searches conducted on databases such as ProQuest, Science Direct, Wiley, and EBSCOhost

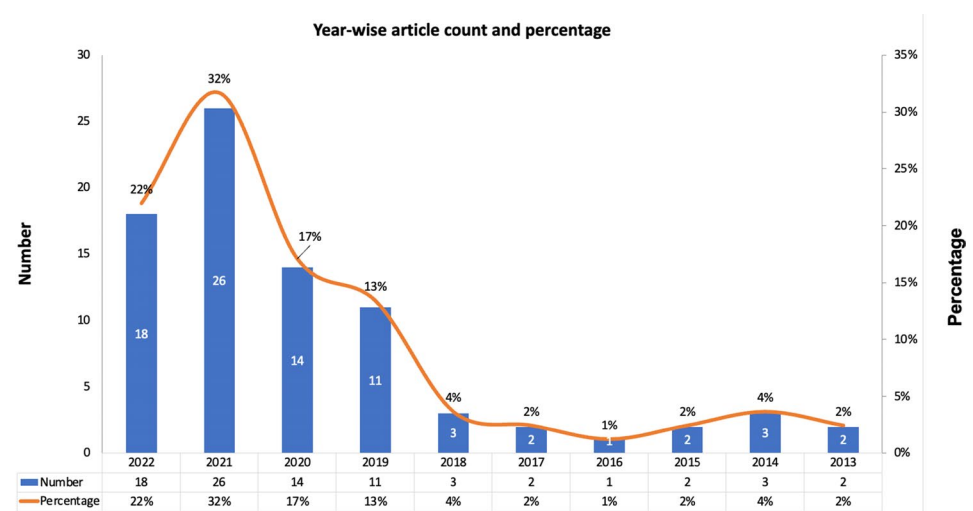


Table 4 List of articles utilized machine learning and deep learning finally selected after following the inclusion and exclusion criteria for systematic literature review process

Technique	Input parameters	Crop	Region	Authors
Polynomial regression, LSTM, and process-based crop models	Temperature and precipitation	Maize	USA, Argentina, Africa, China, India, Indonesia, Canada, Mexico, France, and Brazil	[19]
XGBoost, RF, CNN, and DNN	Phenotype data (flower/seed/pod colour), pubescence density, seed oil content, seed protein content, and seed weight	Soybean	USA	[20]
Linear regression	Precipitation and temperature	Maize	Turkey	[21]
kNN, decision tree, and RF	Nitrogen content ratio, phosphorus content ratio, potassium content ratio in the soil, temperature, relative humidity, soil pH, and rainfall	Maize	India	[22]
Classic statistics and geostatistics	Precipitation, average temperature, average relative precipitation, evapotranspiration, aridity index, content of textural fractions, soil pH, phosphorous, potassium, and zinc in the topsoil (30 cm), soil electrical conductivity, and NDVI, and remotely sensed images	Maize	Hungary	[23]
Back propagation (BP) neural network, SVMs	Phosphorus and optical spectral vegetation indexes	Maize	China	[24]
DayCent ecosystem model Bayesian approach	Daily weather, soil properties, land management, and plant types	Maize	Africa, East Asia, Europe, North America, South America, and South and Southeast Asia	[25]
Novel deep learning adaptive crop model (DACM), SVR, RF, DNN, CNN, and LSTM	Yield data, cropland data layer (CDL), MODIS surface reflectance and a land surface temperature	Soybean	USA	[26]
Univariate prediction model, and multivariate prediction	Crop data (yield and maps) and MODIS data (surface reflectance and phenology data)	Maize and Soybean	USA	[27]
APSIM	Daily weather data (precipitation, air minimum/maximum temperature, solar radiation, and stress degree days (SDD))	Maize	Croatia	[28]
CNN, RF, DBSCAN (Density-based spatial clustering of applications with noise), spectral clustering	Maximum NDVI	Maize	Saudi Arabia	[29]
Ridge regression, kNN, SVR, Gradient boosted decision trees (GBDT)	Weather observations, remote sensing indicators and soil water holding capacity, daily air temperature, daily precipitation, daily evapotranspiration of short vegetation, daily global incoming shortwave radiation, climate water balance, remote sensing, crop areas, irrigated area, elevation, slope, soil water holding capacity, field size, and yield	Maize	Bulgaria, Germany, Spain, France, Hungary, Italy, the Netherlands, Poland, and Romania	[30]
RF and SVMs	Precipitation, minimum air temperature, maximum air temperature, global radiation, wind speed, air humidity, and soil moisture index	Maize	Germany	[31]

Table 4 (continued)

Technique	Input parameters	Crop	Region	Authors
Extremely randomized trees (ERT) model	Yield data, long-term mean of 10 day MODIS NDVI data, precipitation, reference evapotranspiration, and temperature	Maize	Africa	[32]
RF	Yield data, weather data, soil data, and Landsat analysis ready dataset (ARD)	Maize	USA	[33]
Cubic spline regression approach	Yield data and NDVI data	Maize and Soybean	USA	[34]
Multiple regression, computational intelligence, and machine learning (MLP and RBF neural networks)	Phenotypic traits	Maize and Soybean	Brazil	[35]
APSIM and DSSAT	Soil moisture, carbon, nitrogen, and soil temperature	Maize and soybean	USA	[36]
Internet of things (IoT) based crop prediction system/J48, IBk and Random Tree classifiers	Temperature, humidity, soil moisture content, soil pH, soil nutrient content, soil texture, organic matter content, water holding capacity, salinity, and subsoil characteristics	Maize	India	[37]
Extreme learning machine (ELM)	Temperatures, rainfall, air temperature, relative humidity, daily precipitation, and duration of sunshine	Maize	Czechia	[38]
Linear regression, kNN, SVMs, GPR, adaptive boost (AB), and RF	Remote sensing data, temperature, precipitation, fertilizer data, and soil properties data	Maize	USA	[39]
NN	Swth Wdth: The width of the header in feet, Distance: The distance in feet travelled between two data points, Crop Flw (M)(lb/s): The crop flow harvested per second between data points, moisture (%): Crop moisture in %, Speed (mph): The speed of the combine, Prod(ac/h): Combine productivity per hour	Maize	USA	[40]
Lasso regression, ridge regression, elastic net, RF, XGBoost and NN	Weather variables (precipitation, solar radiation, maximum temperature, and minimum temperature), Soil variables (dry bulk density, clay percentage, soil pH, drained upper limit, soil saturated hydraulic conductivity, wilting point, soil organic matter, sand percentage, and saturated volumetric water content)	Maize and Soybean	USA	[41]
Analogy based crop-yield (ABC) forecast scheme	Leaf area index (LAI)	Maize and Soybean	USA	[42]

Table 4 (continued)

Technique	Input parameters	Crop	Region	Authors
RF and SVMs	Leaf area index (LAI), a fraction of absorbed photosynthetically active radiation (FAPAR), Climate data (mean monthly air temperature, minimum monthly air temperature, maximum monthly air temperature, total monthly precipitation, and bioclimatic variables), soil data (nitrogen, organic carbon, soil pH, cation exchange capacity, clay content, silt content, sand content, bulk density), topographic data (digital elevation model, slope, total potential solar radiation, topographic wetness index, wind exposition index), and vegetation data (dry matter productivity, fraction of vegetation cover)	Soybean	Croatia	[43]
RF, SVMs, ANNs, and XGBoost	Daily MODIS datasets soil-adjusted total vegetation index (SATVI), green ratio vegetation index (GRVI), Nadir_B7 (2105–2155 nm), Nadir_B6 (1628–1652 nm), land surface water index (LSWI), NDVI, Nadir_B4 (545–565 nm), and Nadir_B3 (459–479 nm)	Maize	China	[44]
Linear, lasso, ridge, elastic net, and partial least square regression (PLSR)	Vegetation index- Blue green pigment index (BGI), brightness index (BI), excessive green (EXG), excess green minus excess red index (EXGR), green leaf index (GLI), modified green–red index (MGVRI), normalized difference index (NDI), normalized green–blue difference index (NGBDI), red green blue index (RGBVI), visible atmospherically resistant index (VARI), and vegetative (VEG)	Maize	USA	[45]
Parallel Bayesian optimization (PBO) algorithm	Weather data (precipitation, solar radiation, maximum temperature, and minimum temperature), soil variables (dry bulk density, clay percentage, soil pH, drained upper limit, saturated hydraulic conductivity, drained lower limit, organic matter, sand percentage, and saturation), management variables (acres planted at the county level, weekly cumulative percentage of planted and harvested acreages, and nitrogen applications at the county level)	Maize	USA	[46]
RF	Agrometeorological indicators (Precipitation and Temperature), remote sensing vegetation parameters (gross primary production), NDVI and standardized precipitation index	Maize	China	[47]

Table 4 (continued)

Technique	Input parameters	Crop	Region	Authors
Adaptive network-based fuzzy inference system (ANFIS) and multilayer perceptron method Ensembled back propagation networks	Agricultural production yield and losses	Maize and Soybean	Iran	[48]
	Annual average temperature, annual average shortwave radiation, annual total rainfall, slope, soil organic carbon, soil bulk density, soil cation exchange capacity, soil pH, carbonate content, soil moisture, irrigation, crop area, crop yield, nitrogen fertilizer rate, phosphorus fertilizer rate, potassium fertilizer rate, nitrogen-balance, and phosphorus balance	Maize and Soybean	China	[49]
Cubist, RF, XGBoost, and SVMs	Climate data (maximum temperature, mean temperature, minimum temperature, potential evapotranspiration, vapor pressure, precipitation, cloud cover percentage and wet day frequency), satellite data (EVI, NDVI and adjusted vegetation index (SAVI) from MODIS), meteorological indices (growing degree day), extreme degree day (EDD), and the standardized precipitation evapotranspiration index (SPEI)	Maize	China	[50]
	EVI-based forecasts with those based on NDVI, leaf area index (LAI) and fraction of absorbed photosynthetic active radiation (FPAR). MODIS-based subsets of NDVI, EVI, LAI, and FPAR for those 16 years	Maize	USA	[51]
Radiative transfer modelling (RTM), partial-least squares regression (PLSR), and RTM-RF	Leaf nitrogen, chlorophyll, leaf area index (LAI), harvested grain yield, relative humidity, and air temperature	Maize	USA	[52]
ANN, RF, SVR, and GPR	Spatial information about LAI, average leaf inclination angle (ALIA), leaf chlorophyll content, and leaf mass per area	Maize	Germany	[53]
Cubist, XGBoost, stochastic gradient boosting (GBM), SVMs and RF	20 vegetation indices (VI) such as corrected transformed vegetation index (CTVI), difference vegetation index (DVI), global environmental monitoring index (GEMI), and green chlorophyll vegetation index (GCVI)	Soybean	Africa	[54]
GPR	Optical satellites products (enhanced vegetation index (EVI), LAI, and fraction of absorbed photo-synthetically active radiation (FAPAR), microwave satellites products (soil moisture and vegetation optical depth), climatic data (temperature, precipitation, solar radiation, and evaporation)	Maize	United Kingdom and Turkey	[55]

Table 4 (continued)

Technique	Input parameters	Crop	Region	Authors
LASSO regression, ridge regression, SVR, RF, and ANN	Climate variables (monthly mean temperature, precipitation, and vapor pressure deficit (VPD)), remote sensing-based predictors NDVI, EVI, near-infrared reflectance of vegetation (NIRv), and LST data from MODIS	Maize and Soybean	USA	[56]
Lasso regression, SVR, RF, XGBoost, LSTM and CNN	Air temperature, precipitation, vapor pressure deficit, standardized precipitation index (SPI), extreme degree day (EDD), surface reflectance, EVI, green chlorophyll index (GCI), normalized difference water index (NDWI), daytime and nighttime land surface temperature (LST), surface soil moisture, evapotranspiration (ET), potential ET, root zone soil moisture, water stress, palmer drought severity index (PDSI), soil available water content, organic matter, soil nitrogen content, crop progress, harvest area, percentage of harvested area out of county area, percentage of irrigated area out of county area, last 5 year yield mean	Maize	USA	[57]
RF and MLR	Temperature and precipitation	Maize	USA	[58]
DeepCropNet (DCN), LASSO, and RF	Growing degree days, killing degree days, and precipitation	Maize	USA	[59]
PSR (penalized spline regression), C&RT (classification and regression tree), RF, SVMs, and ANNs	Farm size, total labor, soil factors, seed rate, fertilizer, and organic manure	Maize	India	[60]
NN	Soil properties such as micro and macro nutrients	Maize	China	[61]
SVR and RF	MODIS vegetation indices (green NDVI, NDVI, normalized difference moisture index, gross primary production, and fraction of photosynthetically active radiation)	Maize	Kenya	[62]
Cedar river watershed (CRW) model	LAI, weather, topography, land use, soil properties, agricultural management data, nitrate, and nitrogen	Soybean and maize	USA	[63]
RNN and LSTM	Number of days between last freezing temperature and first freezing temperature, sand, silt, clay, organic matter, density, carbonate (CO ₃), water storage, temperature, rainfall, wind speed, and electrical conductivity	Maize	USA	[64]
Linear algorithms (LDA and LR), nonlinear tools- (naive Bayes algorithm, kNN, regression tree algorithm (CART), and SVMs	Rainfall and yield data	Maize	Africa	[65]

Table 4 (continued)

Technique	Input parameters	Crop	Region	Authors
RF	Topographical and soil information (aluminium, phosphorus, potassium, potassium saturation, calcium, calcium saturation, magnesium, magnesium saturation, soil pH, buffer pH, cation exchange capacity, and soil organic matter), and weather variables (monthly average temperature and precipitation)	Maize and Soybean	USA	[66]
Agricultural Production Systems Simulator (APSIM)	Weather conditions, management practices, soil drainage systems, and soil properties (soil nitrate and ammonium)	Maize and Soybean	USA	[67]
RNN and LSTM	Yield data, hourly weather data (temperature, rainfall, wind speed, growing degree days, soil quality data, and soil moisture data)	Maize	USA	[68]
APSIM crop model, LASSO Regression, Ridge Regression, RF, and XGBoost	Crop management, soil management, nitrogen fertilizer management, soil characteristics (texture, soil organic carbon and plant-available water holding capacity), and weather data (temperature and precipitation)	Maize	USA	[69]
DNN	Soil type, land type, soil moisture, temperature, humidity, soil pH, area planted, soil nitrogen, soil phosphorus, soil potassium, rainfall, production, and class label	Maize	India	[70]
DNN and MLR	Climate change (temperature, precipitation, pressure, and circulation), monthly data for cloud cover (%), diurnal temperature range, ground frost frequency, potential evapotranspiration, precipitation, near-surface temperature minimum, near-surface temperature maximum, vapor pressure, and wet day frequency	Soybean	China	[71]
ANN	Climate variables (precipitation, maximum temperature, minimum temperature, and potential evapotranspiration), soil moisture, land cultivated, and NDVI	Maize	Africa	[72]
LSTM-based deep learning model, and GPR	Remote data-MODIS satellite images, NDVI, EVI, the highest average NDVI value, and the average temperature during the growing period	Maize	Africa	[73]
Regularized linear regression and its nonlinear extension -kernel ridge regression	EVI, Vegetation Optical Depth (VOD), and lag between EVI derived from MODIS and VOD	Maize and Soybean	USA	[74]

Table 4 (continued)

Technique	Input parameters	Crop	Region	Authors
SNN (Semi parametric Neural network)	Minimum and maximum air temperature, relative humidity, precipitation, incoming short-wave radiation (sunlight), average wind speed, and growing degree day	Maize	USA	[4]
Multivariate regression	MODIS vegetative index (NDVI, EVI, and NDWI), LAI, leaf angle, soil brightness, and chlorophyll content	Maize and soybean	Africa	[75]
RF, NN, SVMs with radial and linear kernel functions, Gradient Boosting Model (GBM), and Cubist (CU)	Remotely sensed image-derived variables-five soil properties (soil organic matter, cation exchange capacity, magnesium, potassium, and soil pH), yield monitor based maize yield data with multispectral aerial images and topographic data, and yield data	Maize	USA	[76]
MLR, RF, SVMs, kNN, ANNs, and weighted histograms regression	Weather, soil, yield, and regional statistical parameters (yield check, yield difference, planting date, geographic coordinates, temperature, precipitation, solar radiation, cation exchange capacity, soil pH, organic matter, soil class, clay silt, sand, area, and productivity index)	Soybean	USA	[77]
CNN, LSTM, and GPR	Remote sensing data on surface reflectance, land surface temperature, and land cover type	Soybean	USA	[78]
DNN	Daymet (daily surfaces of minimum and maximum temperature, precipitation, humidity, shortwave radiation, and snow water) and MODIS EVI	Maize	USA	[79]
RF	NDVI	Soybean	Brazil	[80]
Neutral networks	NDVI	Maize	Ukraine	[81]
Neutral networks	NDVI	Maize	Italy	[82]
MLR, stepwise linear regression, M5 regression trees, and ANN	Section (farm location where crop was sowed), irrigation water depth applied (mm), solar radiation, rainfall (mm), maximal temperature, minimal temperature, relative humidity in leaf	Maize	Mexico	[83]
Decision tree	Physiological and agronomic features (planting date, stem dry weight, soil type, phosphorus applied, kernel number, kernel weight, season duration (days), soil pH, maximum kernel water content, nitrogen applied, cob dry weight, days to silking, plant density, hybrids type, kernel dry weight, kernel growth rate, duration of the grain filling period, defoliation, leaf dry weight, and potassium applied)	Maize	Iran	[84]

Table 4 (continued)

Technique	Input parameters	Crop	Region	Authors
Hidden Markov model (HMM) based method	NDVI, fractal dimension, and accumulated growing degree days (AGDDs)	Maize	USA	[85]
NN	NDVI	Maize	Italy	[86]

MLR multiple linear regression, *ANN* artificial neural networks, *NN* neural networks, *SVMs* support vector machines, *SVR* support vector regression, *RF* random forest, *kNN* k nearest neighbours, *DNN* deep neural networks, *CNN* convolutional neural networks, *XGBoost* extreme gradient boost, *GPR* gaussian process regression, *LSTM* long short-term neural networks, *RNN* recurrent neural networks, *EVI* enhanced vegetation index, and *NDVI* normalized difference vegetation index

Table 5 List of articles utilized ensemble/hybrid modelling selected after following the inclusion and exclusion criteria for systematic literature review process

Ensemble/hybrid technique	Crop	Input parameters	Region	Authors
Markov Chain Monte Carlo (MCMC) algorithm within a Bayesian framework	Maize	Climate inputs, crop area, crop planting and harvesting dates	USA	[87]
CNN-LSTM	Soybean	Crop growth variables and environment variables, which includes weather data (precipitation and vapor pressure), MODIS land surface temperature (LST) data, and MODIS surface reflectance (SR) data, and historical yield data	CONUS	[88]
CNN-RNN Other single techniques- RF, deep fully connected neural networks (DFNN), and LASSO regression	Maize and soybean	Yield data, management data (weekly cumulative percentage of planted fields), weather variables (precipitation, solar radiation, snow water equivalent, maximum temperature, minimum temperature, and vapor pressure), soil data (wet soil bulk density, dry bulk density, clay percentage, upper limit of plant available water content, lower limit of plant available water content, hydraulic conductivity, organic matter percentage, soil pH, sand percentage, and saturated volumetric water content variables)	USA	[89]
CNN-DNN, CNN-XGBoost, CNN-RNN, and CNN-LSTM Single techniques- XGBoost	Soybean	Weather (precipitation, solar radiation, snow water equivalent, vapor pressure, minimum and maximum temperature), soil (wet soil bulk density, dry bulk density, clay percentage, the upper limit of plant-available water content, lower limit of plant-available water content, hydraulic conductivity, organic matter percentage, soil pH, sand percentage, and saturated volumetric water content), and management data	USA	[90]
Crop modelling + ML (Linear regression, LASSO, LightGBM, RF, and XGBoost)	Maize	Yield data, weather data (daily minimum air temperature, daily maximum air temperature, daily total precipitation, growing degree days, daylight average incident shortwave radiation), soil data (soil organic matter, sand content, clay content, soil pH, soil bulk density, wilting point, field capacity, and saturation point), and management data	USA	[91]
ANNs + RF, and LSTM + RNN + SVMs CARS (competitive adaptive reweighted sampling)-Bayes, SPA (successive projections algorithm)-Bayes, PCA (principal component analysis)-Bayes, CARS-SVMs, SPA-SVMs, PCA-SVMs, CARS-KNN, SPA-KNN, PCA-KNN, CARS-EL, SPA-EL, PCA-EL, CARS-ANN, SPA-ANN, PCA-ANN, MSC (multiplicative scatter correction)-CARS-EL Single techniques- SVMs, kNN, ensemble learning (EL), and ANN	Maize Soybean	Temperature, rainfall, soil pH value, relative humidity, and area Crude protein and crude fat	India China	[92] [93]
SSTNN (spatial-spectral-temporal neural network) (3D convolutional and recurrent neural networks) Single techniques- decision trees, RF, SVMs, LSTM, and 2D CNN	Maize	Image dataset—surface reflectance dataset and MODIS annual land cover dataset, and crop yield dataset	China	[94]
CNN-LF Single techniques- CNN, linear model, a fully connected neural network, RF, and SVMs	Maize	Nitrogen and seed rates prescription maps, elevation map and soil's shallow electro- conductivity. Soil EC measurements texture, bulk density, soil organic carbon, water content, salinity, and cation exchange capacity	USA	[95]

ANN artificial neural networks, SVMs support vector machines, RF random forest, kNN k nearest neighbours, DNN deep neural networks, CNN convolutional neural networks, XGBoost extreme gradient boost, LSTM long short-term neural networks, RNN recurrent neural networks

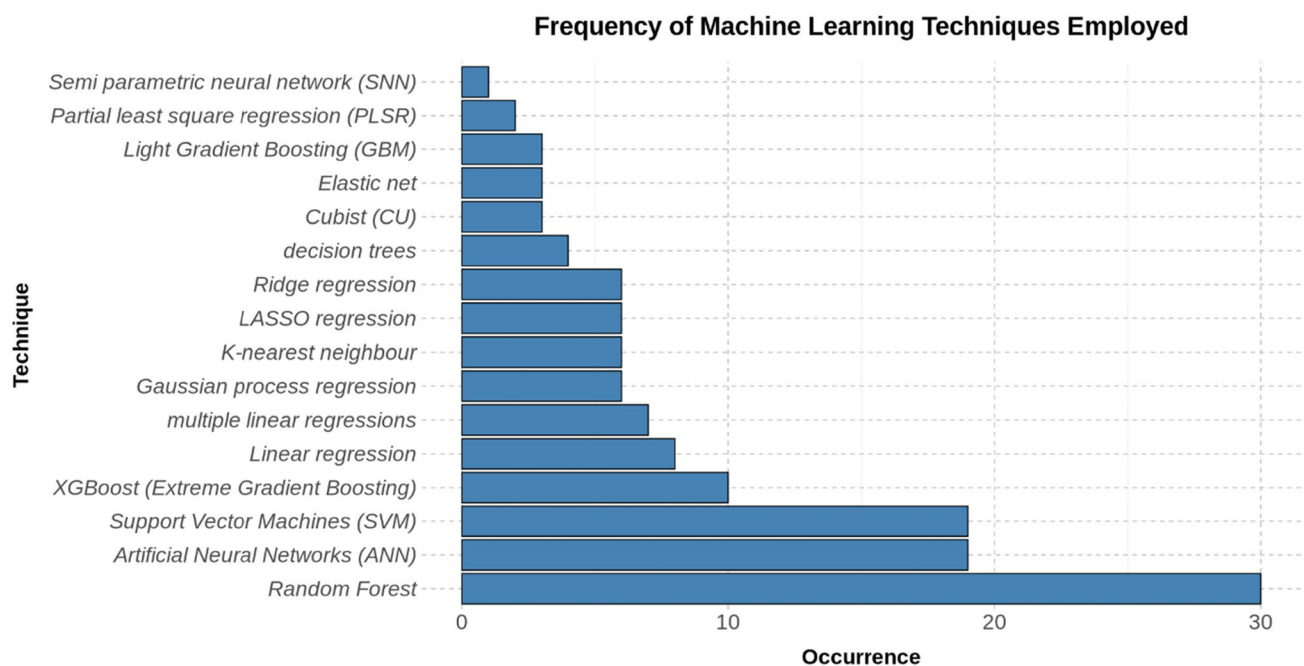


Fig. 4 Frequency of machine learning techniques employed in the literature reviewed. Key features of the most widely used ML models are mentioned in Table S1

qualitative data, which is the case with most of the agricultural and modelling, and (iii) it being a combination of several decision trees, is less prone to overfitting, making it specifically suitable for datasets containing noise [96–98]. The most used DL technique was noted to be the LSTM followed by other widely used techniques such as CNNs, recurrent neural network (RNN), and deep neural network (DNN) (Fig. 5). The LSTM is most extensively used as it has memory cells/gates in its architecture, enabling it to easily decode temporary dependencies that makes it reliable in predicting weather, crop growth, and yields [99]. The advancements in DL frameworks such as PyTorch or TensorFlow has simplified the steps of implementing the LSTM, adding to its large adoption [100]. Also, LSTM's compatibility or ease of getting merged with other models to record temporal/spatial dependencies in data batches, adds to the flexibility of using it at a wider scale [101]. The type of technique to be used is determined by the problem at hand and the nature of the data available [102]. Therefore, it is essential to select appropriate technique that can handle and model the data accurately. Since some papers used more than one technique, total numbers of techniques shown in table exceeds the number of papers (82) considered. This indicates the dominance of multi-technique approach in the study domain where the researchers usually utilize a combination of models to improve prediction accuracy and robustness.

Ranking different models to find the most used ones, as was done for ML and DL, respectively was not possible for ensemble models because every paper presented unique combination of techniques, making it difficult to persistently

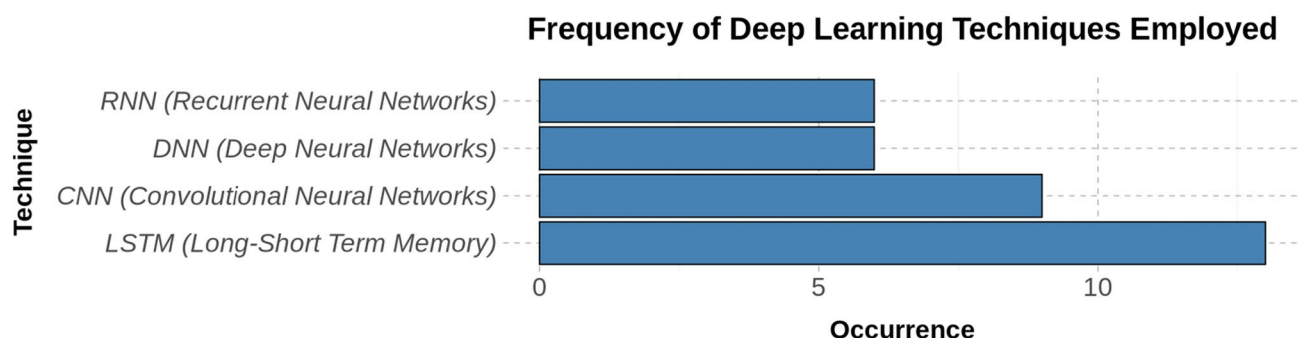


Fig. 5 Frequency of deep learning techniques employed in the literature reviewed. Key features of the most widely used DL models are mentioned in Table S1

determine a set roster of best performing techniques [103]. Hence, for this section, we have highlighted the pros/cons of ensemble models over individual models as per the papers reviewed. For instance, Shahhosseini et al. [91] analysed the impact of merging crop modelling and ML techniques in maize yield prediction in the US and found that proposed algorithm can decrease RMSE of predicted model. Also, they suggested that the proposed models need more hydrological inputs to make improved yield predictions [91]. Another study by Sun et al. [88] proposed a hybrid of CNN and LSTM to predict soybean yield. The prediction accuracy of the proposed hybrid technique CNN-LSTM outperformed the pure CNN or LSTM model. The study of Leng [87] documented the utility of combining Markov chain monte Carlo (MCMC) with Bayesian framework to predict maize yield in USA. Agarwal and Tarar [92] proposed hybrid of ANN with RF and LSTM with RNN and SVMs. The accuracy of two hybrid combinations were compared. The prediction accuracy of hybrid LSTM + RNN + SVMs (97%) proved to be better as compared to the other combination *i.e.* ANN + RF (93%). Khaki et al. [89] merged CNNs and Recurrent Neural Networks (CNN-RNN) for maize and soybean yield prediction and noted the better accuracy of hybrid model. Hence, it implies that different combinations of ML and DL models have been explored by researchers and all combinations of these techniques have been proved more accurate and reliable in terms of performance as compared to the single ML or DL technique. Table 5 presents the hybrid techniques considered by selected studies along with the input parameters and region for the study considered.

R-Q2: What type of input parameters have been most extensively used for predicting maize and soybean yield using ML?

To answer the R-Q2, all the input parameters utilized by ML techniques employed in articles, were analysed, and synthesized. Table 6 is displaying all the used input parameters. As per Table 6, the most used parameters were temperature, precipitation, historical crop yield, NDVI, and soil pH-value, with crop yield as the response variable.

For better understanding of the input parameters, these parameters were further grouped under categories such as soil parameters, crop parameters, climate parameters, nutrient parameter, field management parameters, and remote sensing parameters. The frequency of occurrence of these categories (groups) in the analysed papers is shown in Table 7.

Table 6 Input parameters used for prediction

Input parameters	Occurrence
Temperature	44
Precipitation	30
Historical crop yield	19
NDVI	18
Soil pH-value	16
Organic matter percentage	12
Soil bulk density	11
Rainfall	10
Humidity and relative humidity	10
Enhanced vegetation index (EVI)	10
Solar radiations	9
Surface reflectance	9
Growing degree days	9
Nitrogen	8
Management data	8
Clay content	8
Leaf area index (LAI)	8
Evapotranspiration	7
Area of production	7
Wind speed	6
Cation exchange capacity (CEC)	6
Carbon	5
Fertilizer	4
Potassium	4
Phosphorus	3
Sunshine hours	2

Table 7 Categories of input parameters

Parameters	Occurrence
Climate parameter	46
Remote sensing parameter	33
Soil parameters	26
Crop parameters	18
Nutrient parameter	12
Field management parameter	11

The most used input parameters were climate, remote sensing, and soil parameters (Table 7). The crop, nutrient, and field management parameters coming down in the line implies that these parameters are still in their early phases of getting utilized in the application of ML techniques for corn and soybean yield prediction. The input parameter's grouping category of soil parameters comprise features like soil properties, soil type, soil pH value, soil maps, soil moisture, volumetric water content, soil temperature, organic matter content, salinity, dry bulk density, clay/silt content, hydraulic conductivity, electrical conductivity, cation exchange capacity, water holding capacity, and soil texture. Crop parameters include crop growth, seed variety, plant density, weight, flower/seed/pod colour, seed oil content, seed protein content, seed weight, fraction of vegetation cover, plants type, fraction of absorbed photosynthetically active radiations, and leaf area index. The input parameters grouped under the climate parameters consists of temperature, humidity, evapotranspiration, rainfall, aridity index, shortwave radiations, growing degree days, stress degree days, wind speed, solar radiations, sunshine hours, and air pressure. Nutrient parameters consisted of either the nutrient levels already present in the soil, or the nutrients added to the soils using different fertilizers. These nutrient parameters comprise nitrogen, magnesium, zinc, sulfur, potassium, and phosphorus. Field management parameters consist of elevation of field, field size, topographical wetness index, wind exposition index, fertilizers, irrigation, and other farmers practices. Remote sensing parameters include NDVI, MODIS enhanced vegetation index (MODIS-EVI) and enhanced vegetation index make a category of vegetation index.

R-Q3: What type of accuracy measures and software have been most used for ML and DL techniques for maize and soybean yield prediction?

Among the several measures of evaluating the forecasting accuracy of ML and DL, the most frequently used ones included the R^2 , RAE, and RMSE. Altogether, these three accuracy measures were used in 48% of the cases,

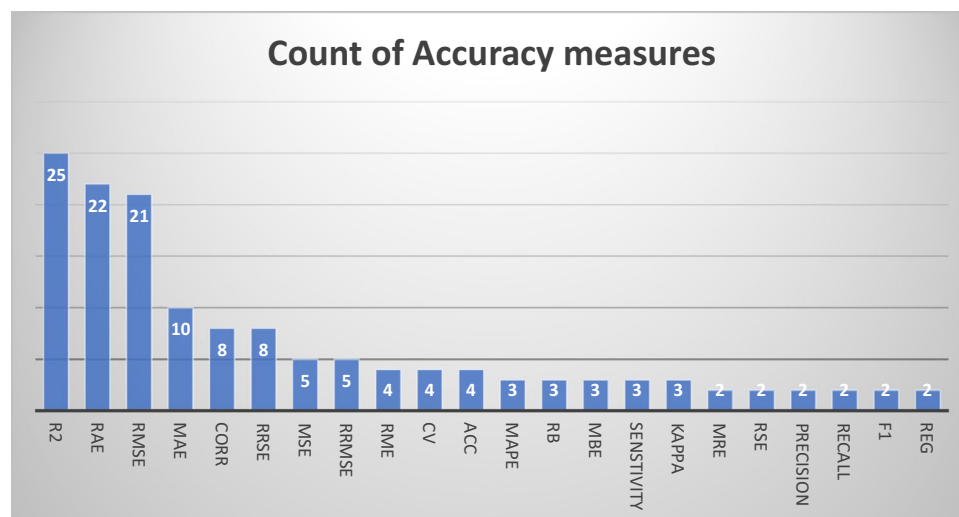


Fig. 6 Count of accuracy measures used in the studies contained in the systematic literature review. R^2 is coefficient of determination, RAE is relative absolute error, RMSE is root mean square error, MAE is mean absolute error, Corr. is correlation coefficient, RRSE is relative root squared error, MSE is mean squared error, RRMSE is relative root mean square error, RME is relative mean error, CV is coefficient of variation, Acc. is accuracy, MAPE is mean absolute percentage error, MBE is mean bias error, Kappa is Cohen's kappa, MRE is mean relative error, RSE is relative squared error, Recall is true positive rate, F1 is F1 score (harmonic mean of precision and recall), and Reg. is regression coefficient. All these accuracy measures are commonly used in prediction models

representing a significant proportion of the total count when all accuracy measures were considered (Fig. 6). Other widely used accuracy measures constitute MAE, correlation coefficient, and relative root square error (RRSE) which collectively comprise 18% of the instances, compared to the total number of times considering all accuracy measures. The detail of consideration of measurement techniques are presented in Appendix 1. Out of the 82 reviewed papers, 23 authors reported the use of readymade software such as Python, MATLAB, Weka, R, and SPSS. Majority of the simulations were performed utilising Python libraries including Python-Scikit and Python-Keras [20, 22, 30, 68, 88].

R-Q4: What is the comparative performance of ML and DL techniques in the field of crop yield prediction?

Table 8 presents a RMSE based performance comparison of various ML, DL, and hybrid prediction techniques employed in reviewed studies. The outperformance of ML over the DL techniques was documented in the study of Gill et al. [20], with higher accuracy noted for RF compared to CNN and DNN. Marko et al. [77] performed comparison of various ML techniques including RF, MLP, ANN, k-NN and SVMs. In this study, RF was noted as the most accurate in generating predictions. Also, this study concluded that the modern ML techniques are better than traditional regression techniques but, due to its non-linear nature, it is hard to judge the correlation between input parameters and output. Among the studies performed comparison of hybrid, ML, and DL techniques, Qiao et al. [94] reported that the DL-based hybrid SSTNN (Spatial-Spectral-Temporal Neural Network) which is a combination of 3D CNN and RNN, performed best in terms of prediction accuracy. Other benchmark techniques tested in this study were RF, SVMs, Decision trees, LSTM, and CNN. The second-best technique was noted to be the single DL technique (CNN), with the least accurate one was ML technique (Decision trees). Similar conclusions were drawn by You et al. [78]. This study had incorporated a Gaussian Process to improve the accuracy of single technique. The hybrid technique CNN-GP was realized to perform best followed by DL technique CNN and then ML technique Decision trees. Better performance of hybrid techniques has also been reported by Khaki et al. [89], who proposed CNN-RNN model and compared the accuracy with RF, DFNN—deep fully connected neural networks and LASSO regression. Oikonomidis et al. [90] compared the performance of XGBoost ML algorithm, CNN-DNN, CNN-XGBoost, CNN-RNN, and CNN-LSTM. The hybrid CNN-DNN outperformed all the techniques.

In addition to RMSE, reviewed literature has used other 21 accuracy measures listed in Fig. 4 to quantify the prediction performances of numerous ML, DL, and hybrid techniques. Based on these 21 accuracy measures, the ANNs were noted as the most accurate technique [40, 49, 53, 60, 61, 76, 83]. Gill et al. [20], Rao et al. [22], Meng et al. [39], Marko et al. [77], Krause et al. [66], and Kenduiywo et al. [62] documented the RF to be the best performer while Gill et al. [20], Geng et al. [44], Kang et al. [57], and Shahhosseini et al. [69] noted XGBoost to be the most accurate technique for yield prediction. Cho and Lee [71] and Kuwata and Shibasaki [79] evaluated DNN as the top-performing technique. Adak et al. [45] and Mateo-Sanchis et al. [74] identified Ridge Regression, while Jiang et al. [68] and Jiang et al. [64] highlighted LSTM. Shekoofa et al. [84] emphasized Decision Trees, and Kenduiywo et al. [62] determined SVR as the most accurate techniques in their respective studies. In addition to evaluating standalone techniques, the accuracy of hybrid techniques has been evaluated using various measures documented in the reviewed literature, encompassing 82 papers. Notably, hybrid techniques that outperformed individual techniques include LSTM-RNN-SVMs, as determined by Agarwal and Tarar [92], MSC-CARS-EL by Zhang et al. [93], CNN-LSTM by Sun et al. [88], CNN-RNN by Khaki et al. [89], and CNN-LF by Barbosa et al. [95] in their respective studies.

R-Q5: What are the challenges in maize and soybean yield prediction research utilizing ML and DL techniques?

For answering the R-Q5 on identifying the challenges in ML and DL research, all 82 papers were thoroughly reviewed to record if they mentioned any limitations/issues in the modelling they performed or if they suggested any future research improvements [7]. Numerous studies reported the problems with the quality and volume of datasets used. These studies mentioned that managing and collecting large sized data was a complex and challenging task in agriculture specifically considering small field size. Quality here meant the data should come from high resolution scales or if geolocation is attached to the datapoints to enable more accurate predictions. Another problem stated in the several publications was the model intricacy, due to the intertwined nature and large set of factors affecting yield which made it difficult for ML models to entirely account for all the causal (crop growth and performance) factors into consideration. Other model-specific problems stated were i) the increased complexity in training-cum-calculative time owing to large data size ii) mandatory requirement of customizing the algorithm for every problem at hand, and iii) designing large data architecture for ensuring high scalability and flexibility. Overall, the studies suggested that more data-sources (with varying resolution scales), data-variety (wider cross-sectional data), and larger time-series (a greater number of years) of data should be incorporated for future testing. Lastly, the reviewed studies suggested the ML techniques to include farm management parameters in future research and generate user-friendly software interface systems to assist farmers in utilizing model's output for on-farm decision making.

Table 8 Prediction performance of various techniques (RMSE)

References/ techniques	ML techniques					DL techniques					Hybrid					SSTNN = 3DCNN + RNN			
	XGBoost	RF	Ridge	MLR	ANN	k-NN	SVMs	DT	Lasso	DFNN	LSTM	CNN	DNN	CNN-DNN	CNN-XGBoost		CNN-LSTM	CNN-RNN	CNN+GP
[20]	6.41	6.32										6.33	6.91						
[89]		26.02							31.30	21.37							17.64		
[89]		12.78							9.49	5.89							4.91		
[77]		10.43		11.25	32.10	14.10	11.04												
[94]		1.37						1.20	1.42		1.12	0.93							0.82
[90]	4.94													0.266	0.346	0.415	0.350		
[78]			7.96					7.73			6.27	5.77	8.32					5.55	5.83

ML machine learning, DL deep learning, XGBoost extreme gradient boost, RF random forest, Ridge ridge regression, MLR multi-linear regression, ANN artificial neural networks, k-NN k nearest neighbours, SVMs support vector regression, DT decision tree, Lasso lasso regression, DFNN deep feed-forward neural network, LSTM long short term memory, CNN convolutional neural network, DNN deep neural network, RNN recurrent neural network, GP gaussian process, SSTNN spatial-spectral-temporal neural network

4 Discussions

General discussion The SLR studies are susceptible to validity risks such as external and construct validity and the reliability which can threaten the trustworthiness of the study findings [7]. The external validity determines the scope to which the study results can be generalized to the settings, sample groups, or framework, outside the situations under which the research is conducted [104]. The construct validity is how accurately the study variables are reflecting the concepts that they are intended to assess [104]. However, both these validity concerns were well responded to in this SLR. The pilot search strategy followed in SLR comprised of a wide-ranging search-string, covering the entirety of the scope of the SLR, consequently retrieving the total 1859 papers initially. To address the reliability concern, the methodology of this SLR is explained in a step-by-step manner with increased clarity considering it can be replicated with ease in future. In case the similar study will be replicated, it may yield different number of papers depending on the personalized perspective. Yet, the chances of getting outcomes different from those of current study are highly improbable.

Discussion relating to search There can always be the chance of missing some valuable research papers. Also, possibilities always exist of getting more research papers by using additional synonyms of keywords by broadening the search scope. Yet, the high number (1859) of research papers retrieved is evidence of all-encompassing search.

Discussion relating to analysis The reason for validity risk could also come from the way how analysis is performed [101]. Notably, not all the 82 papers contained consistent information on (i) type of accuracy measures used, (ii) explanation of features used, (iii) statement of problems, limitations, challenges, or future improvements suggestions. Therefore, the information/data required to address the research problem under question was extracted from fewer than 82 papers. Even though it cannot rectify all the problems, but the other way could be reaching out to the authors for additional or missing information. However, this protocol was not deemed workable for the current study.

R-Q1: Most widely used ML and DL techniques related discussion The ML applications in agriculture initiated in 1980s but they are on rise in crop yield prediction research since 2016 [7, 12]. The ML is even newer in maize and soybean considering 84% of the total research published after 2019 (Fig. 3). This infancy of ML in yield prediction domain along with the vast number of model options adds even more challenges to the model selection task [105]. For example, this SLR itself stated the usage of 87 different ML/DL models in the reviewed literature (Tables 4, 5). In this scenario, the top used ML/DL models identified by the current study can provide a reference point for the future research to compare the effectiveness of new models/techniques in maize and soybean. This will ensure the newer advancements in crop yield prediction research to be evaluated against the verified standards. Another notable aspect is that the Figs. 4 and 5 should be understood carefully considering “most used” doesn’t always mean “best in class” [7]. For example, the linear regression, ranked number five in Fig. 4, normally used as a benchmarking model to assess if the proposed model is superior to the linear regression. Future research encompassing multiple datatypes including weather, remote sensing, and soil could further improve the prediction performance in these identified models. Relying on the footsteps of current study, similar crop-centric model identifications could advance the overall ML research in crop production.

R-Q2: Input parameters related discussion The total 239 features (Tables 4 and 5) used in the reviewed literature were categorized into 26 input parameters (Table 6), which finally grouped into 6 categories (Table 7) to enhance clarity and simplify understanding and analysis. The current findings on top used categories and input parameters are comparable and consistent with the results of Van Klompenburg et al. [7]. The three categories including climate, remote sensing, and soil parameters coming up in the line indicates the high adoption of ML techniques in these areas. For these three categories, the altogether frequency of 105 out of 146 implies that the 72% of the total ML-yield prediction research was centred on these three areas. Similar can be understood by the frequency distribution of the input parameters *i.e.*, temperature, precipitation, historical crop yield, NDVI, and soil pH-value which is 127 out of 283 or 45%. This is attributable to the feasibility in accessing many online sources for large datasets related to climate and soil properties and the in-built capabilities of remote sensing systems to generate large volume of data [106, 107]. These large datasets are mandatory for running ML models. Contrarily, such large datasets are difficult to obtain in field/greenhouse research settings which serves as the primary data source for field management and nutrient parameters [108]. Consequently, ML techniques adoption in field and nutrient management practices is still not on par with that of climate, soil, and remote sensing research. However, the increasing penetration of sensing technology, data fusion, and geostatistical tools in field research is building a platform for merging ML in field level research for more practical applications [108].

R-Q3: Accuracy measures and software related discussion There were total 22 accuracy measures utilized for assessing model performances, including RMSE, MAE, RRSE, MAPE, and R^2 . Notably, researchers usually inclined

towards using the R^2 , as evidenced by studies such as Yin et al. [19], Maitah et al. [38], Meng et al. [39], and Geng et al. [44], whereas RAE was preferred as per research conducted by Viswanath et al. [37], Sun et al. [88], Kussul et al. [81], Kaneko et al. [73], and Zhu et al. [26], and RMSE was widely used, as recognized by Gill et al. [20], Nosratabadi et al. [48], Liang et al. [49], Kang et al. [57], and Cho and Lee et al. [71]. Considering ML in its infancy in crop yield prediction research, knowing about the most used accuracy measures and the software listing could be helpful for the future researchers in selecting the best suitable tool for the problem at hand. Additionally, the most used software has their wide user communities, support systems, and documentations to help users in trouble shooting purposed if needed.

R-Q4: Comparative performance of ML and DL techniques related discussion The DL techniques have shown higher accuracy in comparison to ML techniques, as evaluated by various studies [78, 89, 90, 92, 94]. Contrarily, dissimilar viewpoints showed by other studies [20], that documented the ML technique of RF to outperform DL techniques. Likewise, Kang et al. [57] concluded that the XGBoost algorithm shown better performance compared to deep neural networks such as LSTM and CNN. Hybrid models are robust in unifying the first principles with data-driven frameworks by coalescing the multiple ML techniques [109]. They are adept in integrating the domain mastery with real-world data, and hence, their search-ability and reasoning methodology is best-in-class (in most cases) [110]. Hybrid model's architectural setup works on the principle of merging the rule-induction methods and case-based reasoning approach [111]. This improves predictive accuracy and extrapolation abilities, offering enhanced efficiency in solving both regression and classification complex problems [112]. Despite this, hybrid models are less frequently used due to various challenges associated with their adoption. Some of the challenges include (i) more and better computing infrastructure and more cycle/run time is required for a hybrid model execution in comparison to the individual or simple models [113], (ii) more specialized efforts and higher complexity is required to estimate hybrid models [114], (iii) Also, sometimes the type of problem at hand is the deciding factor for selecting the model type, for instance, for the problems requiring identification of heterogenous preferences, hybrid modelling is better while for higher predictive accuracy, the simpler models are preferred [114].

R-Q5: Challenges related discussion: The challenges were reported in this SLR solely based on what was mentioned in the reviewed papers. There can be the chances that some additional challenges did exist but were not reported in the papers. However, this SLR reported major challenges with obtaining high quantity/quality data, model complexity tackling, and inclusion of farm management factors. However, data quantity and quality are integral parts of data-training protocol of ML systems that decide their accuracy [115]. High quality data in agriculture is again challenging as the extremeness in variability coming from changes in geography, climate, soil, and other management factors that changes with geography [116]. However, possibilities do exist for generating a synthetic data utilizing internet of things or satellite imagery to train automated systems for improving crop recommendation process [117, 118]. Variational autoencoder and generative adversarial networks are the techniques which were noted successful in tackling the problem of limited data in agriculture by generating synthetic data useable for crop recommendation systems [118, 119]. Regional biases, insufficient longitudinal data, or the challenges of integrating multiple data sources present another set of challenges with the ML applications. Paudel et al. [30] pointed out the issue of spatial biases encountered while combining the smaller geographical unit's data to create larger ones, passing on errors inherited from geospatial land maps or recorded data. Inadequate longitudinal data causes issues with analysing temporal variations while combining multiple data (*i.e.*, weather, soil, or remotely sensed vegetation) sources adds to the complications necessitating meticulous pre-processing and feature selection procedures [120]. The security/privacy issues attached to the collection, usage, and storage of the crop production data adds to additional hurdles for ML research for harnessing high quality/volume datasets for model-training protocols [121]. Merging ML-based outputs to create end user or farmer-friendly software interface systems is a work-in-progress in its initial phase. To effectively fasten this progress in making the ML system's usability, more cross-disciplinary research involving data scientists, agricultural economists, crop modelers, farmers, and other stakeholders is needed in the yield prediction domain [122]. Also, incorporating the explainable artificial intelligence into the existing models could enhance the transparency of the recommendations generated by ML-assisted smart agriculture systems [123].

5 Conclusions

This study reviewed 82 published papers to identify the most used ML/DL techniques, input parameters, and accuracy measures, as well as the challenges in maize and soybean yield prediction research. The analysed studies covered 32 diverse geographical regions, used 87 different ML/DL techniques with 239 varied features, and assessed 22 different

accuracy measures. The results deduced that no precise conclusion on which model is best can be made but the results clearly separated out the specific ML and DL models that were used more than others. Specifically, the RF, ANNs, SVMs, and XG-Boost were noted as the most used ML algorithms. Most applied DL techniques include LSTM neural network and CNNs. In these models, the most used performance assessment measures were noted as the R^2 , RAE, RMSE, and MAE. This study revealed the temperature, precipitation, NDVI, and soil pH to be the most primary features utilized in ML by yield prediction research. This SLR reported major challenges with obtaining high quantity/quality data, model complexity tackling, and inclusion of farm management factors in the yield prediction process. Also, there is a need to fasten the process of merging ML-based outputs to create farmer-friendly software interface systems, enhancing the ML system's usability, by conducting more cross-disciplinary research involving data scientists, agricultural economists, crop modelers, farmers, and other stakeholders. Similar research needs to be conducted in other important crops for exploring most usable ML and DL models, features, and accuracy measures as well as identifying most encountered challenges.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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Appendix 1

See Table 9.

Table 9 Accuracy measures

Authors	RMSE	MAE	R ²	Corr	RME	MSE	MPE	RMAE	RRSE	MAPE	RAE	RB	MBE	MRE	MARE	RSE	RRMSE	CV	Std Dev	Precision	Recall	f1	Acc	Reg	Sensitivity	Kappa	PE	
[19]	1		1								1	1	1															
[28]																												
[20]	1																											
[21]																		1	1									
[22]																				1	1	1	1					
[23]				1														1										
[24]																							1	1				
[25]	1											1																
[35]			1																									
[36]																												
[37]		1									1						1											
[87]																								1				
[38]	1		1																									
[92]																				1	1	1	1					
[39]	1		1																									
[40]																						1						
		1	1								1						1	1										
[42]																										1		
[43]																									1			
[44]	1	1	1																									
[45]	1	1	1																									
[46]	1	1	1														1											
[47]			1	1												1												
[48]	1	1	1																									
[49]	1			1	1																							
[91]	1																											
[50]			1																									
[57]	1	1		1						1		1																
[58]																		1										
[60]	1		1																									
[61]	1	1																										
				1						1																		
[62]	1		1																									
[63]						1																						
[64]		1								1																		
[69]	1		1										1				1											
[70]			1			1																						
[71]	1	1		1									1															
[72]	1		1																					1				
[93]	1			1																								

Table 9 (continued)

Authors	RMSE	MAE	R ²	Corr	RME	MSE	MPE	RMAE	RRSE	MAPE	RAE	RB	MBE	MRE	MARE	RSE	RRMSE	CV	Std Dev	Precision	Recall	f1	Acc	Reg	Sensitivity	Kappa	PE
[88]											1															1	
[4]		1	1		1																						
[75]		1	1																			1					
[77]		1	1			1																					
[80]																											
[81]											1																
[82]		1	1																								
[83]		1	1			1	1																				
[85]																											
[86]																											
[78]									1																		
[76]											1																
[73]			1								1																
[89]		1	1																								
[65]																						1					
[26]		1							1		1																
[27]		1							1		1																
[52]		1	1	1							1							1									
[56]											1																
[94]									1		1																
[53]																											
[54]	1					1					1																
[34]											1																
[55]											1																
[30]	1								1		1																
[32]									1																		
[95]																											
[51]									1		1															1	
[33]									1		1																
[74]											1																
[66]											1																
[67]											1																
[68]																1											
[90]											1																
[79]	1	1	1			1																					
Total	21	10	25	8	4	5	1	1	8	3	22	3	3	2	1	2	5	4	1	2	2	2	4	2	3	3	1

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